

In [1]: `import pandas as pd`

In [2]: `movies = pd.read_csv(r"C:\Users\suras\OneDrive\Desktop\Movies.csv", sep=',')
print(type(movies))
movies.head(20)`

<class 'pandas.core.frame.DataFrame'>

Out[2]:

	movieId	title	genres
0	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
1	2	Jumanji (1995)	Adventure Children Fantasy
2	3	Grumpier Old Men (1995)	Comedy Romance
3	4	Waiting to Exhale (1995)	Comedy Drama Romance
4	5	Father of the Bride Part II (1995)	Comedy
5	6	Heat (1995)	Action Crime Thriller
6	7	Sabrina (1995)	Comedy Romance
7	8	Tom and Huck (1995)	Adventure Children
8	9	Sudden Death (1995)	Action
9	10	GoldenEye (1995)	Action Adventure Thriller
10	11	American President, The (1995)	Comedy Drama Romance
11	12	Dracula: Dead and Loving It (1995)	Comedy Horror
12	13	Balto (1995)	Adventure Animation Children
13	14	Nixon (1995)	Drama
14	15	Cutthroat Island (1995)	Action Adventure Romance
15	16	Casino (1995)	Crime Drama
16	17	Sense and Sensibility (1995)	Drama Romance
17	18	Four Rooms (1995)	Comedy
18	19	Ace Ventura: When Nature Calls (1995)	Comedy
19	20	Money Train (1995)	Action Comedy Crime Drama Thriller

In [3]: `tags = pd.read_csv(r"C:\Users\suras\OneDrive\Desktop>tag.csv", sep=',')
tags.head()`

```
Out[3]:
```

	userId	movieId	tag	timestamp
0	18	4141	Mark Waters	2009-04-24 18:19:40
1	65	208	dark hero	2013-05-10 01:41:18
2	65	353	dark hero	2013-05-10 01:41:19
3	65	521	noir thriller	2013-05-10 01:39:43
4	65	592	dark hero	2013-05-10 01:41:18

```
In [8]: ratings = pd.read_csv(r"C:\Users\suras\OneDrive\Desktop\ratings.csv", sep=',')
ratings.head
```

```
Out[8]: <bound method NDFrame.head of
```

	userId	movieId	rating	timestamp
0	1	2	3.5	02-04-2005 23:53
1	1	29	3.5	02-04-2005 23:31
2	1	32	3.5	02-04-2005 23:33
3	1	47	3.5	02-04-2005 23:32
4	1	50	3.5	02-04-2005 23:29
...	...	...	...	...
1048570	7120	168	5.0	02-04-2007 19:44
1048571	7120	253	4.0	02-04-2007 19:30
1048572	7120	260	5.0	02-04-2007 19:27
1048573	7120	261	4.0	02-04-2007 19:49
1048574	7120	266	3.5	02-04-2007 19:34

```
[1048575 rows x 4 columns]>
```

```
In [9]: del ratings['timestamp']
del tags['timestamp']
```

```
In [10]: tags.head()
```

```
Out[10]:
```

	userId	movieId	tag
0	18	4141	Mark Waters
1	65	208	dark hero
2	65	353	dark hero
3	65	521	noir thriller
4	65	592	dark hero

Data Structures:

```
In [11]: tags.iloc[0]
```

```
Out[11]:  userId      18
         movieId    4141
         tag        Mark Waters
         Name: 0, dtype: object
```

```
In [13]: row_0=tags.iloc[0]
```

```
In [15]: row_0
```

```
Out[15]:  userId      18
         movieId    4141
         tag        Mark Waters
         Name: 0, dtype: object
```

```
In [16]: row_0.index
```

```
Out[16]: Index(['userId', 'movieId', 'tag'], dtype='object')
```

```
In [19]: row_0['userId']
```

```
Out[19]: 18
```

```
In [20]: 'rating' in row_0
```

```
Out[20]: False
```

```
In [22]: row_0.name
```

```
Out[22]: 0
```

```
In [23]: row_0 = row_0.rename('firstRow')
         row_0.name
```

```
Out[23]: 'firstRow'
```

```
In [24]: tags.head()
```

```
Out[24]:
```

	userId	movieId	tag
0	18	4141	Mark Waters
1	65	208	dark hero
2	65	353	dark hero
3	65	521	noir thriller
4	65	592	dark hero

Data Frames

```
In [25]: tags.head()
```

Out[25]:

	userId	movieId	tag
0	18	4141	Mark Waters
1	65	208	dark hero
2	65	353	dark hero
3	65	521	noir thriller
4	65	592	dark hero

In [26]: `tags.index`

Out[26]: RangeIndex(start=0, stop=465564, step=1)

In [27]: `tags.columns`

Out[27]: Index(['userId', 'movieId', 'tag'], dtype='object')

In [29]: `tags.iloc[[1,34,877]]`

Out[29]:

	userId	movieId	tag
1	65	208	dark hero
34	65	58652	Stephen Chow
877	359	36517	'There's A Lotta These At Used Book Stores...H...

Descriptive Statistics

In [31]: `ratings['rating'].describe()`

Out[31]:

count	1.048575e+06
mean	3.529272e+00
std	1.051919e+00
min	5.000000e-01
25%	3.000000e+00
50%	4.000000e+00
75%	4.000000e+00
max	5.000000e+00
Name: rating, dtype: float64	

In [32]: `ratings.describe()`

```
Out[32]:
```

	userId	movieId	rating
count	1.048575e+06	1.048575e+06	1.048575e+06
mean	3.527086e+03	8.648988e+03	3.529272e+00
std	2.018424e+03	1.910014e+04	1.051919e+00
min	1.000000e+00	1.000000e+00	5.000000e-01
25%	1.813000e+03	9.030000e+02	3.000000e+00
50%	3.540000e+03	2.143000e+03	4.000000e+00
75%	5.233000e+03	4.641000e+03	4.000000e+00
max	7.120000e+03	1.306420e+05	5.000000e+00

```
In [33]: ratings['rating'].mean()
```

```
Out[33]: 3.5292716305462175
```

```
In [34]: ratings.mean()
```

```
Out[34]:
```

userId	3527.086123
movieId	8648.988281
rating	3.529272
dtype:	float64

```
In [35]: ratings['rating'].min()
```

```
Out[35]: 0.5
```

```
In [36]: ratings['rating'].max()
```

```
Out[36]: 5.0
```

```
In [37]: ratings['rating'].std()
```

```
Out[37]: 1.0519187535891295
```

```
In [38]: ratings['rating'].mode()
```

```
Out[38]:
```

0	4.0
---	-----

Name: rating, dtype: float64

```
In [39]: ratings.corr()
```

Out[39]:

	userId	movieId	rating
userId	1.000000	-0.002837	0.017105
movieId	-0.002837	1.000000	0.002550
rating	0.017105	0.002550	1.000000

```
In [40]: filter1 = ratings['rating'] > 10
print(filter1)
filter1.any()
```

```
0      False
1      False
2      False
3      False
4      False
...
1048570 False
1048571 False
1048572 False
1048573 False
1048574 False
Name: rating, Length: 1048575, dtype: bool
```

Out[40]: False

```
In [41]: filter2 = ratings['rating'] > 0
filter2.all()
```

Out[41]: True

Data Cleaning: Handling Missing Data

```
In [43]: movies.shape
```

Out[43]: (27278, 3)

```
In [44]: movies.isnull().any().any()
```

Out[44]: False

```
In [45]: ratings.shape
```

Out[45]: (1048575, 3)

```
In [46]: ratings.isnull().any().any()
```

Out[46]: False

```
In [47]: tags.shape
```

Out[47]: (465564, 3)

```
In [48]: tags.isnull().any().any()
```

Out[48]: True

```
In [49]: tags=tags.dropna()
```

```
In [50]: tags.isnull().any().any()
```

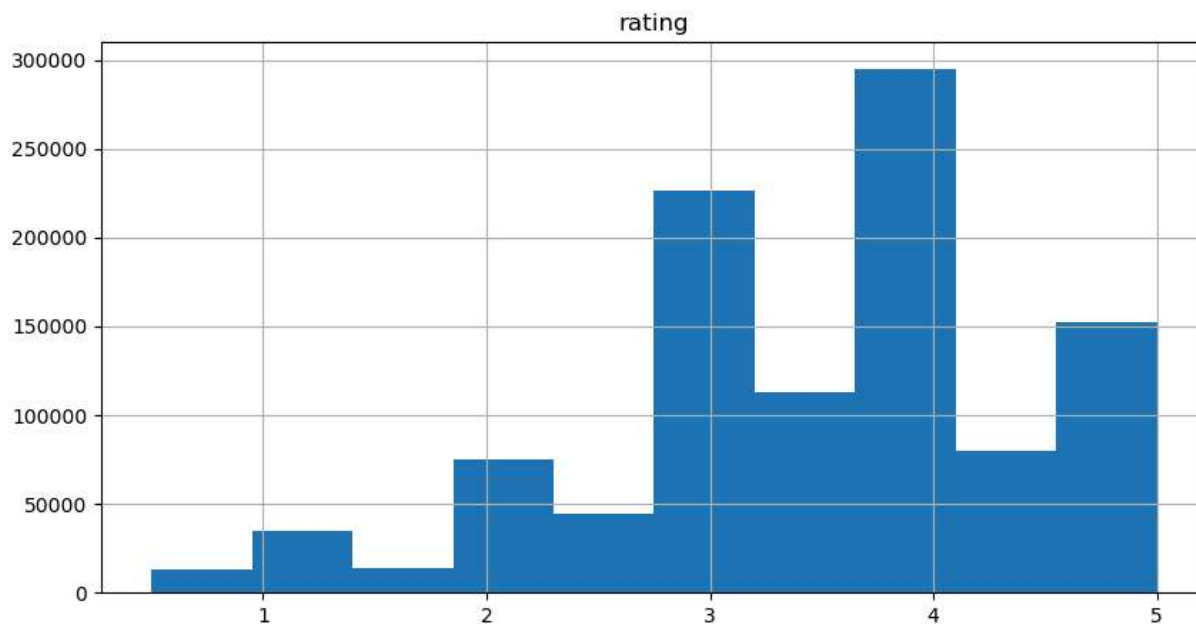
Out[50]: False

```
In [51]: tags.shape
```

Out[51]: (465548, 3)

Data Visualization

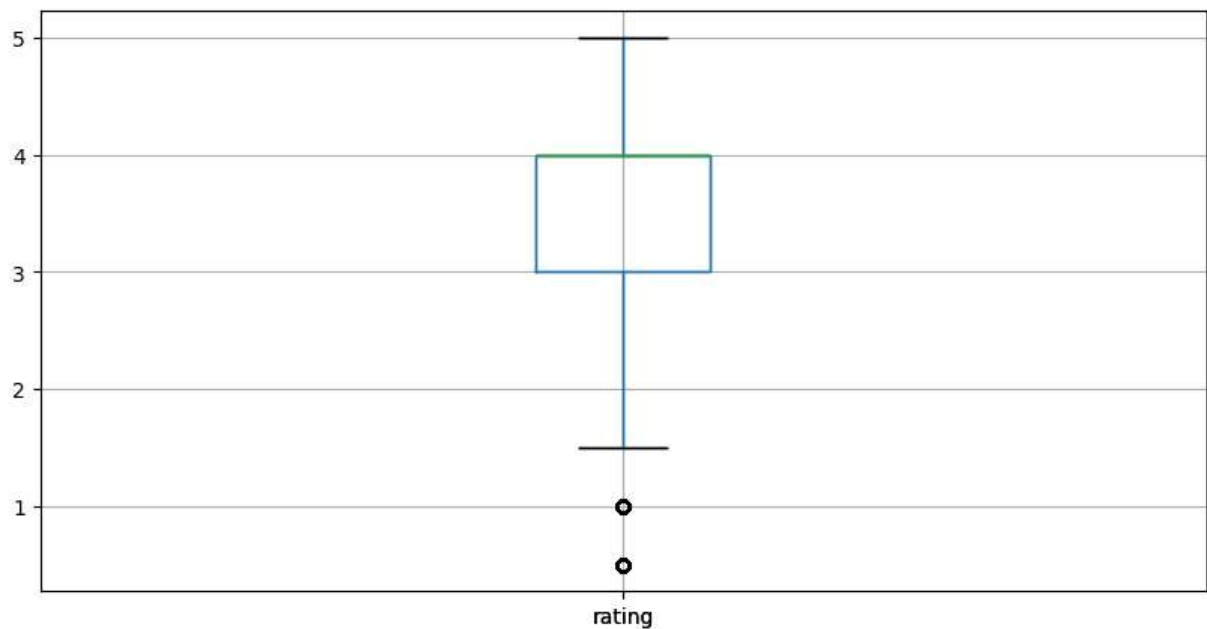
```
In [60]: import matplotlib.pyplot as plt
%matplotlib inline
ratings.hist(column='rating', figsize=(10,5))
plt.show()
```



```
In [62]: ratings.boxplot(column='rating', figsize=(10,5))
```

Out[62]: <Axes: >

```
In [63]: plt.show()
```



Slicing Out Columns

In [64]: `tags['tag'].head()`

Out[64]:

0	Mark Waters
1	dark hero
2	dark hero
3	noir thriller
4	dark hero

Name: tag, dtype: object

In [65]: `movies[['title', 'genres']].head()`

Out[65]:

	title	genres
0	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
1	Jumanji (1995)	Adventure Children Fantasy
2	Grumpier Old Men (1995)	Comedy Romance
3	Waiting to Exhale (1995)	Comedy Drama Romance
4	Father of the Bride Part II (1995)	Comedy

In [66]: `ratings[-10:]`

Out[66]:

	userId	movieId	rating
1048565	7120	141	5.0
1048566	7120	151	5.0
1048567	7120	153	0.5
1048568	7120	161	4.0
1048569	7120	163	4.5
1048570	7120	168	5.0
1048571	7120	253	4.0
1048572	7120	260	5.0
1048573	7120	261	4.0
1048574	7120	266	3.5

```
In [67]: ratings[20:]
```

Out[67]:

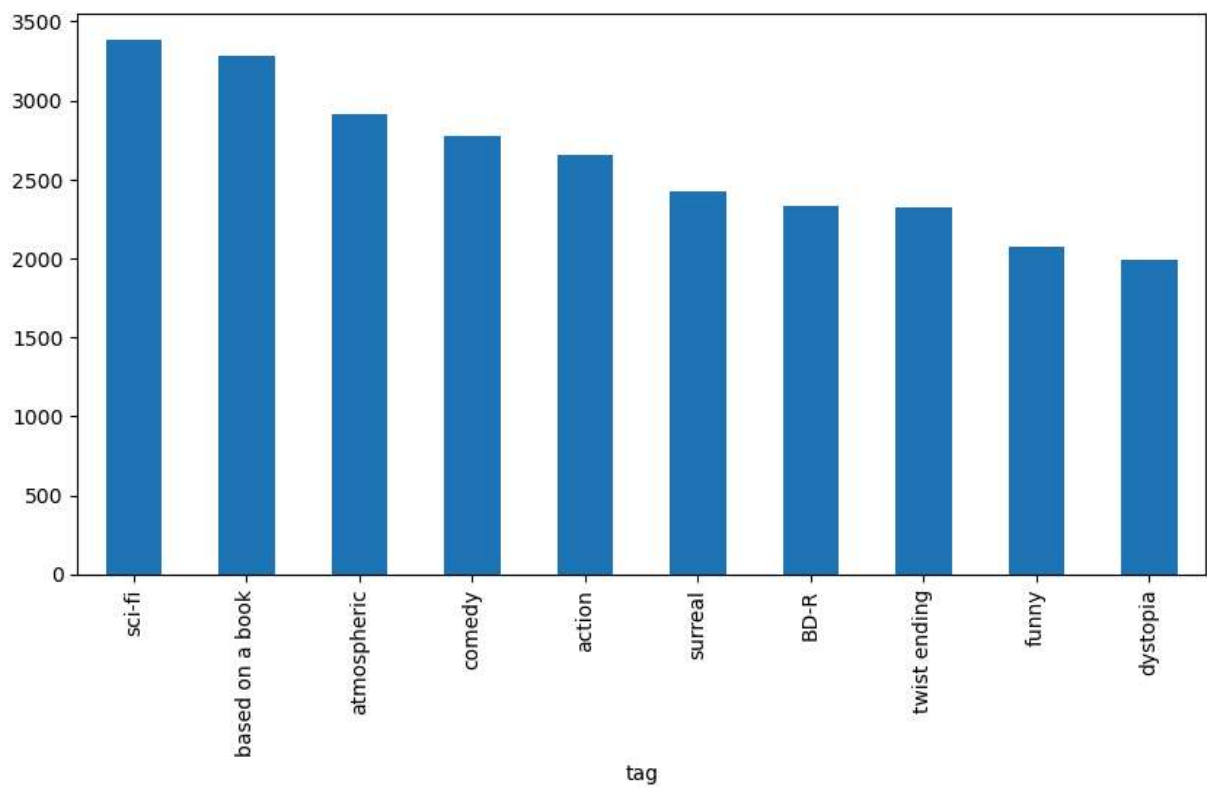
	userId	movieId	rating
20	1	924	3.5
21	1	1009	3.5
22	1	1036	4.0
23	1	1079	4.0
24	1	1080	3.5
...	...	...	...
1048570	7120	168	5.0
1048571	7120	253	4.0
1048572	7120	260	5.0
1048573	7120	261	4.0
1048574	7120	266	3.5

1048555 rows × 3 columns

```
In [68]: tag_counts = tags['tag'].value_counts()
tag_counts[-10:]
```

```
Out[68]: tag
missing child      1
Ron Moore          1
Citizen Kane       1
mullet            1
biker gang         1
Paul Adelstein     1
the wig            1
killer fish        1
genetically modified monsters  1
topless scene      1
Name: count, dtype: int64
```

```
In [70]: tag_counts[:10].plot(kind='bar', figsize=(10,5))
plt.show()
```



```
In [ ]:
```